Chapter 20

Line Integrals and Work

Accumulating scalar quantities and vector forces along curves

Chapter overview

A definite integral accumulates a quantity along an interval. A double integral accumulates a quantity over a region. A triple integral accumulates a quantity through a solid. A **line integral** accumulates a quantity along a curve.

Line integrals are the first place where the geometry of a path becomes part of the integral. A curve can bend, twist, move through changing density, or pass through a force field whose direction changes from point to point. The value of the integral may depend not only on the endpoints, but also on the route taken between them.

This chapter develops two closely related ideas:

1. **Scalar line integrals**, written $\int_C f\,ds$, which accumulate a scalar field along a curve.
2. **Vector line integrals**, written $\int_C F \cdot dr$, which measure the work done by a vector field along an oriented curve.

The second kind is one of the central objects of vector calculus. It connects geometry, physics, conservative fields, Green's theorem, Stokes' theorem, and the dynamics of optimization algorithms.

The main idea

A line integral is a limit of small contributions along a curve. A curve is chopped into short pieces. On each piece, the field is approximately constant and the curve is approximately straight. The line integral is the limiting sum of all these small contributions.

Learning goals

After studying this chapter, you should be able to:

1. define and compute scalar line integrals with respect to arc length;
2. compute line integrals with respect to x, y, and z;
3. compute vector line integrals using a parameterization of the curve;
4. interpret $\int_C F \cdot dr$ as work done by a force field;
5. understand orientation, path dependence, and additivity over piecewise curves;
6. approximate line integrals numerically from sampled points;

7. connect line integrals to motion, optimization paths, probability flows, and machine-learning loss landscapes.

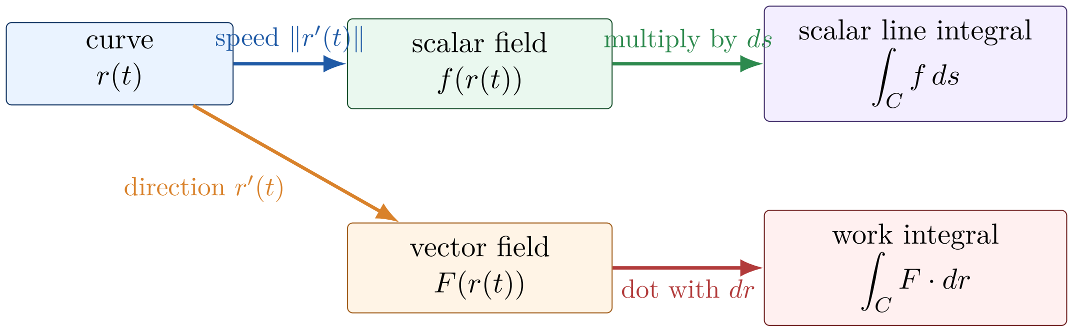

Figure 20.1: Two line integrals use the same path but accumulate different local quantities. Scalar line integrals use length; vector line integrals use oriented displacement.

20.1 Oriented curves and parameterizations

A curve C in the plane or space is usually described by a vector-valued function

$$r(t) = \langle x(t), y(t)\rangle, \qquad a \le t \le b,$$

or

$$r(t) = \langle x(t), y(t), z(t)\rangle, \qquad a \le t \le b.$$

The direction in which t increases gives the **orientation** of the curve. The same geometric curve can have the opposite orientation if we traverse it in reverse.

The velocity vector is $r'(t) = \langle x'(t), y'(t)\rangle$ in the plane, or $r'(t) = \langle x'(t), y'(t), z'(t)\rangle$ in space. Its magnitude is the speed along the curve. The arc length differential is

$$ds = \|r'(t)\|\ dt.$$

For a space curve,

$$\|r'(t)\| = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}.$$

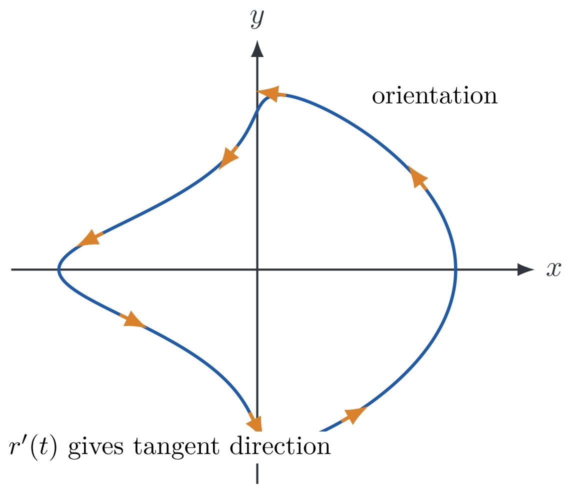

Figure 20.2: An oriented curve with tangent vectors. The arrows show the direction in which the parameter increases.

20.2 Scalar line integrals with respect to arc length

Suppose $f(x, y)$ is a scalar field defined on a curve C. The scalar line integral of f along C with respect to arc length is

$$\int_C f(x, y)\, ds.$$

If C is parameterized by $r(t) = \langle x(t), y(t)\rangle$ for $a \le t \le b$, then

$$\int_C f(x, y)\, ds = \int_a^b f(x(t), y(t))\sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt.$$

For a space curve $r(t) = \langle x(t), y(t), z(t)\rangle$,

$$\int_C f(x, y, z)\, ds = \int_a^b f(x(t), y(t), z(t))\sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\, dt.$$

The factor $ds = \|r'(t)\|\, dt$ converts the parameter interval into actual distance traveled along the curve.

Scalar line integrals do not depend on orientation

Because ds measures length, reversing the orientation of the same curve does not change $\int_C f\, ds$. The curve is traversed in the opposite direction, but the same small lengths are added.

Example 1: scalar line integral over a quarter circle

Evaluate

$$\int_C (3 - xy^2)\, ds,$$

where C is the first-quadrant part of the unit circle $x^2 + y^2 = 1$. Use

$$r(t) = \langle \cos t, \sin t\rangle, \qquad 0 \le t \le \frac{\pi}{2}.$$

Then

$$r'(t) = \langle -\sin t, \cos t \rangle, \qquad \|r'(t)\| = 1.$$

Therefore

$$\begin{aligned}\int_C (3 - xy^2)\,ds &= \int_0^{\pi/2} \left(3 - \cos t \sin^2 t\right)\,dt \\ &= \frac{3\pi}{2} - \frac{1}{3}.\end{aligned}$$

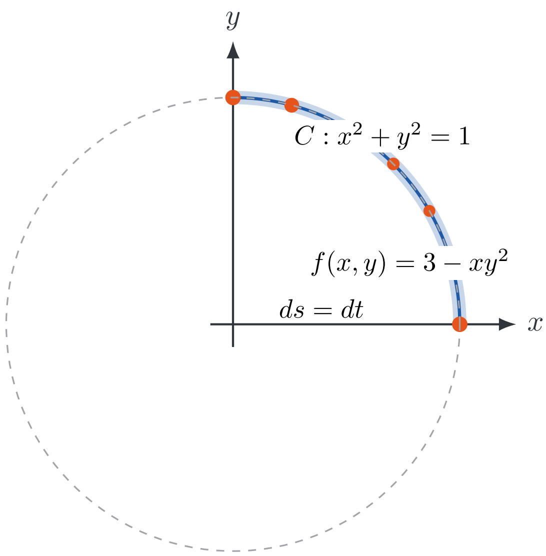

Figure 20.3: A scalar field sampled along a quarter circle. A scalar line integral accumulates field values weighted by arc length. Larger dots indicate larger sampled field values.

20.3 Mass of a wire

A major interpretation of a scalar line integral is the mass of a thin wire. Suppose a wire has shape C and density $\rho(x, y, z)$ measured in mass per unit length. Then the mass is

$$M = \int_C \rho(x, y, z)\,ds.$$

If the density is constant, this becomes density times length. If the density changes from point to point, the line integral accumulates the varying density along the wire.

Example 2: mass of a helical wire

Let C be the helix

$$r(t) = \langle \sin t, \cos t, t \rangle, \qquad 0 \le t \le \pi.$$

Suppose the density is $\rho(x, y, z) = 2x \sin z$. Then $\rho(r(t)) = 2\sin^2 t$. Also

$$r'(t) = \langle \cos t, -\sin t, 1 \rangle, \qquad \|r'(t)\| = \sqrt{2}.$$

Thus

$$\int_C 2x \sin z\,ds = \int_0^{\pi} 2\sin^2 t \sqrt{2}\,dt = \pi\sqrt{2}.$$

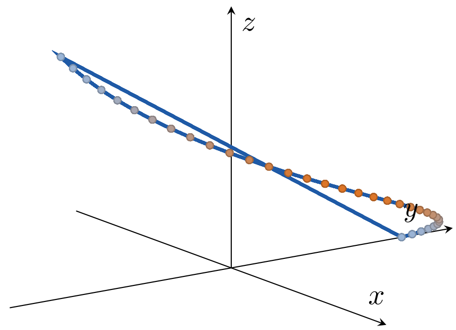

Figure 20.4: A helix with scalar density sampled along the curve. The scalar line integral gives the mass of a wire with this density.

20.4 Line integrals with respect to coordinates

Line integrals can also be taken with respect to coordinates. In the plane, we may see

$$\int_C f(x,y)\,dx, \qquad \int_C g(x,y)\,dy.$$

If C is parameterized by $x = x(t)$ and $y = y(t)$, then

$$\int_C f(x,y)\,dx = \int_a^b f(x(t),y(t))x'(t)\,dt,$$

and

$$\int_C g(x,y)\,dy = \int_a^b g(x(t),y(t))y'(t)\,dt.$$

In space, similarly,

$$\int_C f(x,y,z)\,dz = \int_a^b f(x(t),y(t),z(t))z'(t)\,dt.$$

Unlike ds, the coordinate differentials dx, dy, and dz can be positive or negative. Therefore coordinate line integrals depend on orientation.

ds versus dx, dy, and dz

The symbol ds is a length element and is always nonnegative. The symbols dx, dy, and dz are signed coordinate changes. Reversing the orientation of a curve leaves $\int_C f\,ds$ unchanged, but changes the sign of $\int_C f\,dx$, $\int_C f\,dy$, and $\int_C f\,dz$.

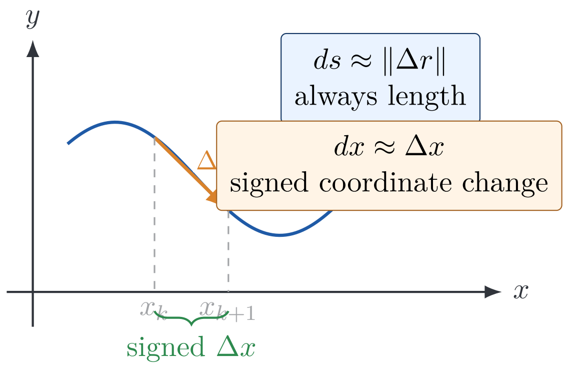

Figure 20.5: The arc length element ds measures actual distance along the curve, while dx measures signed horizontal change.

20.5 Vector line integrals and work

Suppose an object is displaced by a vector d while a constant force F acts on it. The work done is

$$W = F \cdot d.$$

If the force varies from point to point and the object moves along a curve C, we chop the curve into small displacement vectors Δr. On each small piece, the work is approximately $F \cdot \Delta r$. Adding these contributions and taking a limit gives the vector line integral

$$\int_C F \cdot dr.$$

If $F(x, y) = \langle P(x, y), Q(x, y)\rangle$ and $r(t) = \langle x(t), y(t)\rangle$, then

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t)\, dt.$$

Equivalently,

$$\int_C F \cdot dr = \int_C P\, dx + Q\, dy.$$

In space, if $F(x, y, z) = \langle P, Q, R\rangle$, then

$$\int_C F \cdot dr = \int_C P\, dx + Q\, dy + R\, dz.$$

The computation recipe

To compute $\int_C F \cdot dr$:

1. parameterize the curve as $r(t)$;
2. compute $r'(t)$;
3. substitute the parameterization into the field to get $F(r(t))$;
4. take the dot product $F(r(t)) \cdot r'(t)$;

5. integrate with respect to t over the parameter interval.

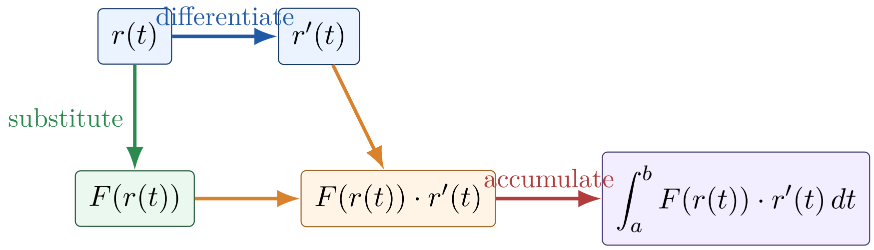

Figure 20.6: The standard workflow for a vector line integral.

Example 3: work along a circular arc

Let

$$F(x, y) = \langle y^2, -xy \rangle,$$

and let C be the curve

$$r(t) = \langle \sin t, \cos t \rangle, \qquad 0 \le t \le \frac{\pi}{2}.$$

Then $r'(t) = \langle \cos t, -\sin t \rangle$. Substituting into the vector field gives

$$F(r(t)) = \langle \cos^2 t, -\sin t \cos t \rangle.$$

The dot product is

$$F(r(t)) \cdot r'(t) = \cos^3 t + \sin^2 t \cos t = \cos t.$$

Therefore

$$\int_C F \cdot dr = \int_0^{\pi/2} \cos t\,dt = 1.$$

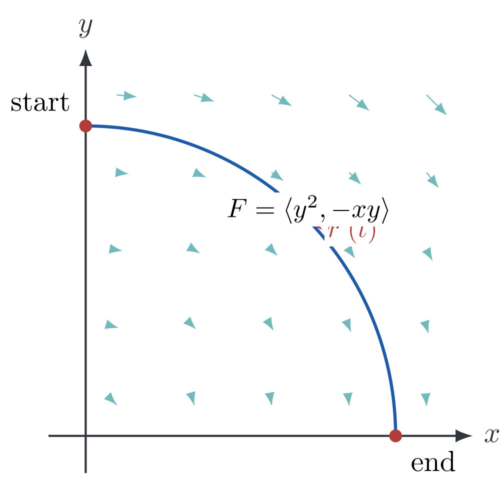

Figure 20.7: Work along a circular arc in a vector field. The integrand is the tangential component of the field times speed.

20.6 Orientation matters for vector line integrals

Let $-C$ denote the same curve as C but with the opposite orientation. Then

$$\int_{-C} F \cdot dr = -\int_C F \cdot dr.$$

This agrees with the work interpretation. Pushing an object forward along a path may do positive work; moving the same object backward through the same field reverses the displacement direction and reverses the sign of the work.

Example 4: reversing a path

For the field and curve in Example 3,

$$\int_C F \cdot dr = 1.$$

If the same arc is traversed from $(1, 0)$ back to $(0, 1)$, then

$$\int_{-C} F \cdot dr = -1.$$

This sign change is one of the most common sources of mistakes in vector line integrals.

20.7 Piecewise smooth curves

Many useful curves are not given by one formula. They are built from several smooth pieces. If

$$C = C_1 \cup C_2 \cup \cdots \cup C_m$$

with compatible orientations, then

$$\int_C F \cdot dr = \int_{C_1} F \cdot dr + \int_{C_2} F \cdot dr + \cdots + \int_{C_m} F \cdot dr.$$

The same additivity holds for scalar line integrals.

Example 5: a piecewise line integral in space

Evaluate

$$\int_C y\,dx + z\,dy + x\,dz,$$

where $C = C_1 \cup C_2$, C_1 is the line segment from $(3, 4, 0)$ to $(3, 4, 5)$, and C_2 is the line segment from $(3, 4, 5)$ to $(2, 0, 0)$.
For C_1, use $r_1(t) = \langle 3, 4, t\rangle$, $0 \le t \le 5$. Then $dx = 0$, $dy = 0$, and $dz = dt$, so

$$\int_{C_1} y\,dx + z\,dy + x\,dz = \int_0^5 3\,dt = 15.$$

For C_2, use $r_2(s) = \langle 3-s, 4-4s, 5-5s\rangle$, $0 \le s \le 1$. Then $dx = -ds$, $dy = -4ds$, and $dz = -5ds$. Therefore

$$\int_{C_2} y\,dx + z\,dy + x\,dz = \int_0^1 [(4-4s)(-1) + (5-5s)(-4) + (3-s)(-5)]\,ds$$
$$= \int_0^1 (-39 + 29s)\,ds = -\frac{49}{2}.$$

Thus

$$\int_C y\,dx + z\,dy + x\,dz = 15 - \frac{49}{2} = -\frac{19}{2}.$$

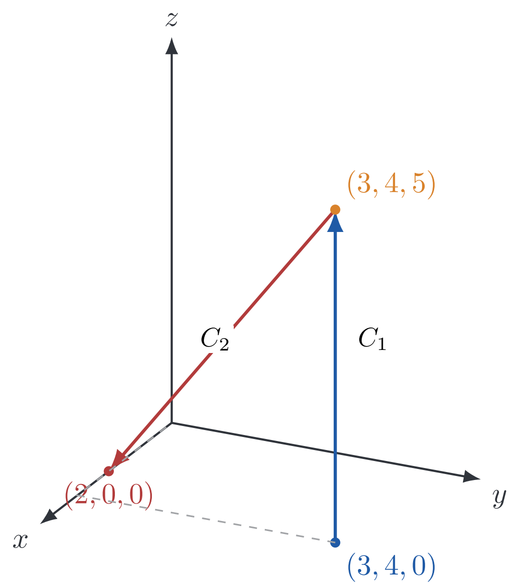

Figure 20.8: A piecewise curve in space. Line integrals over piecewise curves are computed one segment at a time.

20.8 Path dependence

A major difference between line integrals and ordinary definite integrals is that path matters. Two different curves with the same starting and ending points can produce different line integrals.

Example 6: same endpoints, different paths

Let $F(x, y) = \langle y, 0\rangle$. We compute the work from $(0, 0)$ to $(1, 1)$ along two paths.
Path A: straight line. Let $r_A(t) = \langle t, t\rangle$, $0 \le t \le 1$. Then $F(r_A(t)) = \langle t, 0\rangle$ and $r'_A(t) = \langle 1, 1\rangle$. Therefore

$$\int_{C_A} F \cdot dr = \int_0^1 t\,dt = \frac{1}{2}.$$

Path B: horizontal then vertical. First go from $(0, 0)$ to $(1, 0)$, then from $(1, 0)$ to $(1, 1)$. On the horizontal segment, $y = 0$, so $F = \langle 0, 0\rangle$. On the vertical segment, $dx = 0$, and

$F \cdot dr = y\,dx = 0$. Hence

$$\int_{C_B} F \cdot dr = 0.$$

The endpoints are the same, but the work is different. This field is path-dependent.

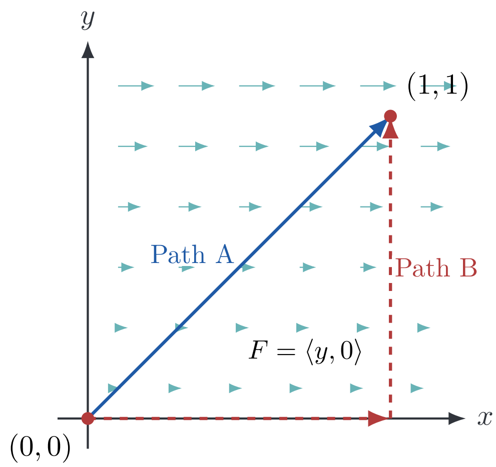

Figure 20.9: Two paths with the same endpoints can produce different vector line integrals.

20.9 A three-dimensional work integral

Let

$$F(x, y, z) = \langle xy, yz, zx \rangle$$

and let C be parameterized by

$$r(t) = \langle t, t^2, t^3 \rangle, \qquad 0 \le t \le 1.$$

Then $r'(t) = \langle 1, 2t, 3t^2 \rangle$. Substituting into F gives

$$F(r(t)) = \langle t^3, t^5, t^4 \rangle.$$

Thus

$$F(r(t)) \cdot r'(t) = t^3 + 2t^6 + 3t^6 = t^3 + 5t^6.$$

Therefore

$$\int_C F \cdot dr = \int_0^1 (t^3 + 5t^6)\,dt = \frac{1}{4} + \frac{5}{7} = \frac{27}{28}.$$

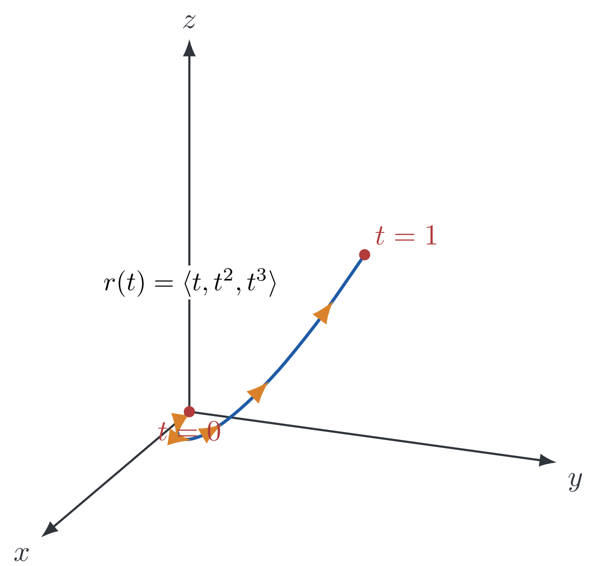

Figure 20.10: A space curve used for a three-dimensional work integral, with tangent directions shown along the path.

20.10 Numerical approximation of vector line integrals

If a curve is sampled at points

$$r_0, r_1, \ldots, r_N,$$

then a numerical approximation to the work integral is

$$\int_C F \cdot dr \approx \sum_{k=0}^{N-1} F\left(\frac{r_k + r_{k+1}}{2}\right) \cdot (r_{k+1} - r_k).$$

This is the midpoint rule along a curve. It is useful when the curve is known only from data, simulation output, robot trajectories, or numerical solutions of differential equations.

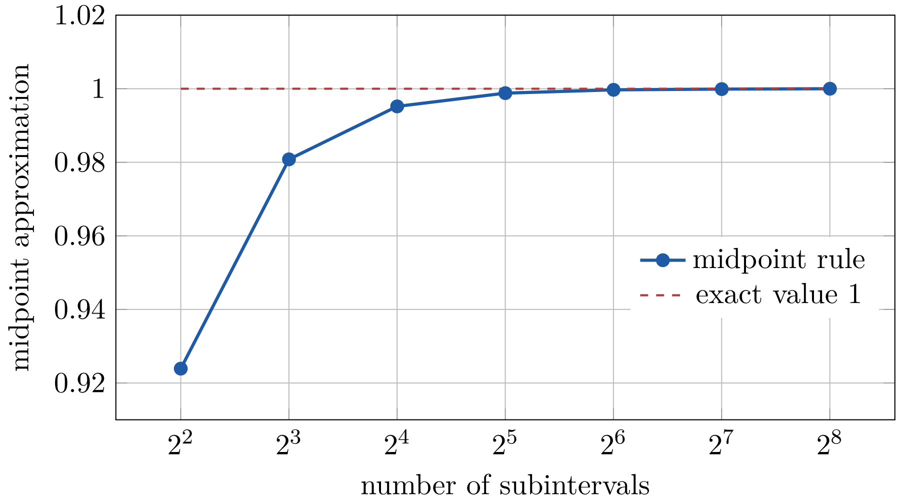

Figure 20.11: Numerical midpoint approximations converge to the vector line integral as the curve sampling becomes finer.

20.11 Line integrals along optimization paths

In machine learning and statistics, we often study a loss function $L(\theta)$, where θ is a vector of parameters. The gradient $\nabla L(\theta)$ is a vector field on parameter space. An optimization algorithm produces a path

$$\theta_0, \theta_1, \theta_2, \ldots$$

through that space.

The work-like quantity

$$\int_C \nabla L \cdot d\theta$$

measures accumulated change of the loss along the path. For a smooth path, the chain rule gives

$$\frac{d}{dt} L(\theta(t)) = \nabla L(\theta(t)) \cdot \theta'(t),$$

so line integrals of gradients are closely tied to change in objective values. This idea becomes the Fundamental Theorem for Line Integrals in the next chapter.

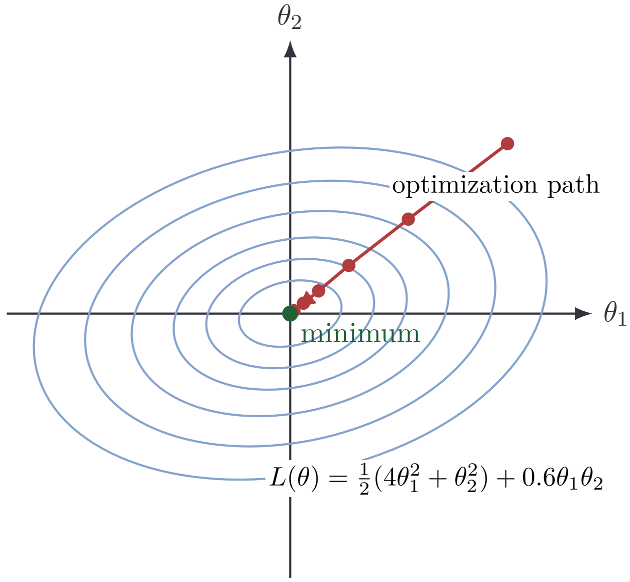

Figure 20.12: A path through a quadratic loss landscape. Line integrals can measure accumulated change along paths in parameter space.

20.12 Work, circulation, and flow

The vector line integral $\int_C F \cdot dr$ measures the component of the field tangent to the curve. If the field points mostly along the direction of motion, the integral is positive. If it points mostly opposite the direction of motion, the integral is negative. If the field is mostly perpendicular to the motion, the contribution is small.

For a closed curve, the integral

$$\oint_C F \cdot dr$$

is called the **circulation** of F around C. Circulation measures how much the vector field tends to push around the loop. It is a central idea behind Green's theorem and Stokes' theorem.

Example 7: circulation of a rotational field

Let $F(x, y) = \langle -y, x\rangle$. On the unit circle with counterclockwise orientation,

$$r(t) = \langle \cos t, \sin t\rangle, \qquad 0 \le t \le 2\pi.$$

Then

$$F(r(t)) = \langle -\sin t, \cos t\rangle, \qquad r'(t) = \langle -\sin t, \cos t\rangle.$$

Thus $F(r(t)) \cdot r'(t) = 1$, so

$$\oint_C F \cdot dr = \int_0^{2\pi} 1\, dt = 2\pi.$$

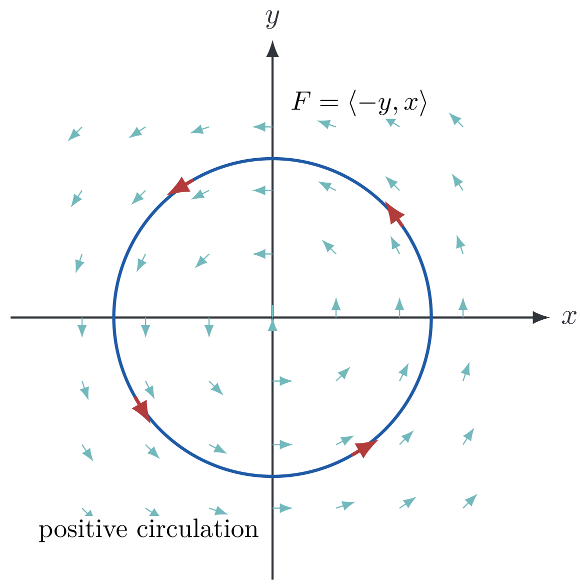

Figure 20.13: Positive circulation around the unit circle for the rotational field $F = \langle -y, x\rangle$.

20.13 Common mistakes

Mistake 1: forgetting $\|r'(t)\|$ in scalar line integrals

For scalar line integrals, the formula is

$$\int_C f\,ds = \int_a^b f(r(t))\,\|r'(t)\|\,dt.$$

The factor $\|r'(t)\|$ is essential unless the parameterization has unit speed.

Mistake 2: using $\|r'(t)\|$ in vector line integrals

For vector line integrals, the formula is

$$\int_C F\cdot dr = \int_a^b F(r(t))\cdot r'(t)\,dt.$$

Do not replace $r'(t)$ by $\|r'(t)\|$. The direction of motion matters.

Mistake 3: ignoring orientation

Vector line integrals depend on orientation. Reversing the path changes the sign.

20.14 Chapter summary

A line integral accumulates quantities along curves.

For scalar fields,

$$\int_C f\,ds = \int_a^b f(r(t))\,\|r'(t)\|\,dt.$$

This measures total mass, charge, cost, or exposure along a curve when f is a density per unit length.

For vector fields,

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t)\,dt.$$

This measures work done by a force field or accumulated tangential flow along an oriented curve. In coordinates,

$$\int_C F \cdot dr = \int_C P\,dx + Q\,dy$$

in the plane, and

$$\int_C F \cdot dr = \int_C P\,dx + Q\,dy + R\,dz$$

in space.

The essential themes are parameterization, orientation, tangential components, and path dependence. The next chapter studies the special vector fields for which line integrals depend only on endpoints.

20.15 Exercises

Conceptual exercises

1. Explain in your own words the difference between $\int_C f\,ds$ and $\int_C F \cdot dr$.
2. Why does reversing the orientation of a curve not change $\int_C f\,ds$?
3. Why does reversing the orientation of a curve change the sign of $\int_C F \cdot dr$?
4. What does it mean for a vector field to do positive work along a curve?
5. Give a physical interpretation of $\int_C \rho\,ds$ when ρ is a density along a wire.
6. Give a data-science interpretation of integrating a cost or risk function along a trajectory.

Computational exercises

7. Evaluate $\int_C (x+y)\,ds$, where C is the line segment from $(0,0)$ to $(2,1)$.
8. Evaluate $\int_C x^2\,ds$, where C is the unit circle parameterized counterclockwise.
9. Evaluate $\int_C y\,dx + x\,dy$, where C is the line segment from $(0,0)$ to $(1,2)$.
10. Evaluate $\int_C F \cdot dr$ for $F(x,y) = \langle x, y\rangle$ and $r(t) = \langle \cos t, \sin t\rangle$, $0 \le t \le 2\pi$.
11. Evaluate $\int_C F \cdot dr$ for $F(x,y) = \langle -y, x\rangle$ and $r(t) = \langle 2\cos t, 2\sin t\rangle$, $0 \le t \le 2\pi$.
12. Evaluate $\int_C F \cdot dr$ for $F(x,y,z) = \langle z, x, y\rangle$ and $r(t) = \langle t, t^2, t^3\rangle$, $0 \le t \le 1$.
13. Evaluate $\int_C z\,ds$ for the helix $r(t) = \langle \cos t, \sin t, t\rangle$, $0 \le t \le 2\pi$.

14. Let $F(x, y) = \langle y, 0\rangle$. Compute the work from $(0, 0)$ to $(1, 1)$ along the curve $y = x^2$.
15. Let $F(x, y) = \langle 0, x\rangle$. Compute the work from $(0, 0)$ to $(1, 1)$ along the straight line and along the path that goes vertically first and then horizontally.

Computational lab exercises

16. Write a function that approximates $\int_C F \cdot dr$ from sampled curve points.
17. Use numerical approximation to estimate the work done by $F(x, y) = \langle \sin y, \cos x\rangle$ along $r(t) = \langle t, t^2\rangle$, $0 \le t \le 1$.
18. Compare the numerical approximation from Exercise 17 for $N = 10, 20, 40, 80, 160$ subintervals.
19. Plot the vector field $F(x, y) = \langle -y, x\rangle$ and three closed curves. Estimate the circulation around each.
20. Generate a random polygonal path from $(0, 0)$ to $(1, 1)$ and estimate $\int_C \langle y, 0\rangle \cdot dr$.

Challenge exercises

21. Find a vector field F and two paths from A to B such that the work along one path is positive and the work along the other path is negative.
22. Let $F(x, y) = \langle ax + by, cx + dy\rangle$. Determine conditions on a, b, c, d so that $\int_C F \cdot dr$ has the same value for every path from $(0, 0)$ to $(1, 1)$.
23. For $F(x, y) = \langle -y, x\rangle$, compute the circulation around a circle of radius a.
24. Let C be any circle centered at the origin, oriented counterclockwise. Explain why $F(x, y) = \langle x, y\rangle$ has zero circulation around C.
25. Suppose a curve is reparameterized by a smooth increasing change of variables. Prove that $\int_C F \cdot dr$ is unchanged.
26. Suppose a curve is reparameterized by a smooth decreasing change of variables. Prove that $\int_C F \cdot dr$ changes sign.